

Abstract

Poisson and Jacobi structures play a fundamental role in the geometric description of many systems arising in classical mechanics. In most cases, the corresponding bivector field is induced by a non-degenerate 2-covariant tensor field. In this poster, we present a unified framework for constructing the associated brackets by studying such tensor fields in detail. More specifically, under suitable assumptions on the tensor, we derive a formula for computing the Schouten–Nijenhuis bracket of the associated bivector field in terms of the curvature of a certain distribution and the exterior derivative of a differential form. This expression provides the obstruction to the existence of Poisson and Jacobi structures. We use this to recover the classical brackets of symplectic, locally conformally symplectic, cosymplectic, contact, and cocontact geometries, and to characterize when *fat bundles* and *almost cosymplectic structures of order p* determine a Jacobi bracket.

1. Introduction

Symplectic geometry revolutionized the study of classical mechanics, enabling differential-geometric methods for Hamiltonian systems (Darboux’s theorem, the Liouville–Arnold theorem, KAM theory). Its success inspired other geometric structures: *contact* geometry (time-dependent, dissipative and thermodynamic systems), *cosymplectic* structures (non-autonomous mechanics), *locally conformally symplectic* manifolds (non-conservative systems), etc... These structures also appear in modern topology, e.g. knot theory and low-dimensional topology.

Recently, some of the authors introduced in [5] a general framework for geometries defined by a pair (τ, ω) of a 1-form and a 2-form (the so-called *doublets*). Most standard structures can be seen as doublets whose *class* (the codimension of $\text{Ker } \tau \cap \text{Ker } \omega$) equals $\dim M$. Equivalently, in this case, the 2-covariant tensor $B = \omega + \tau \otimes \tau$ is non-degenerate.

In all these contexts a bracket of functions is essential. By Lichnerowicz [2, 3], a **Poisson** structure is a bivector $\Lambda \in \mathfrak{X}^2(M)$ with $[\Lambda, \Lambda] = 0$, and a **Jacobi** structure is a pair (Λ, E) with $[\Lambda, \Lambda] = 2E \wedge \Lambda$ and $\mathcal{L}_E \Lambda = 0$; the latter defines a local Lie algebra on $\mathcal{C}^\infty(M)$ in the sense of Kirillov [4]. Here $[\cdot, \cdot]$ is the Schouten–Nijenhuis bracket.

In this work [1], we take the bundle isomorphism $\flat: TM \rightarrow T^*M$ as the core datum, identify it with the left contraction of a (generally neither symmetric nor skew-symmetric) 2-covariant tensor B , and study when the skew-symmetric part of B^{-1} completes to a Poisson or Jacobi bivector. Under suitable conditions, we can provide an *intrinsic, coordinate-free* formula for the Schouten–Nijenhuis bracket.

2. Preliminaries on multilinear algebra

A 2-covariant tensor field $B \in \Gamma(T^*M \otimes T^*M)$ splits canonically as $B = \omega + S$ into its skew-symmetric and symmetric parts, $\omega = \frac{1}{2}(B - tB)$, $S = \frac{1}{2}(B + tB)$. We say that B is *non-degenerate* if its left contraction $\bar{B}: TM \rightarrow T^*M$, $\langle \bar{B}(u), v \rangle = B(u, v)$, is an isomorphism. Right contraction gives the transpose $t\bar{B} = tB$. The induced bivector field is

$$\Lambda(\alpha, \beta) = -\omega(\bar{B}^{-1}(\alpha), \bar{B}^{-1}(\beta)), \quad \Lambda = -(\Lambda^2 t\bar{B}^{-1})(\omega).$$

Definition (asymmetry)
The asymmetry of B is the endomorphism $J = t\bar{B}^{-1} \circ \bar{B}: TM \rightarrow TM$. It satisfies $B(Ju, Jv) = B(u, v)$ and $v \in \text{Ker } \omega \iff J(v) = v, \quad v \in \text{Ker } S \iff J(v) = -v.$ Hence $\text{Ker } \omega$ and $\text{Ker } S$ are always in direct sum, and $TM = \text{Ker } \omega \oplus \text{Ker } S \iff J^2 = \text{Id}$.

Remark. *Symplectic vector spaces* ($J = -\text{Id}$) and *inner products* ($J = \text{Id}$, and B positive-definite) are two extremal cases. When $\dim M$ is even and (ω, g) is a Kähler structure, the asymmetry of $B = g + \omega$ is precisely the induced complex structure J .

For non-degenerate B , the inverse tensor $B^{-1} \in \Gamma(TM \otimes TM)$ (with $\widehat{B^{-1}} = \bar{B}^{-1}$) has skew-symmetric part the bivector $\Lambda = -(\Lambda^2 tB^{-1})(\omega)$ and symmetric part $\Sigma = (\otimes^2 tB^{-1})(S)$. Dually,

$$TM = \text{Ker } \omega \oplus \text{Ker } S \iff T^*M = \text{Ker } \Lambda \oplus \text{Ker } \Sigma.$$

Orthogonality. For a subspace W and a 2-covariant tensor T , we define $W^{\perp, T} = \{v : T(v, w) = 0 \ \forall w \in W\}$. Writing $W^{\perp} = W^{\perp, B}$ and ${}^\perp W = W^{\perp, tB}$, one has that $\dim W^\perp = \dim {}^\perp W = \text{codim } W$ and

$${}^\perp(W^\perp) = W, \quad W^{\perp\perp} = J^{-1}(W), \quad B|_W \text{ non-degenerate} \iff TM = W \oplus W^\perp.$$

Direct-sum decomposition. When $TM = \mathcal{H} \oplus \mathcal{V}$ with $\mathcal{H} := \text{Ker } S$, $\mathcal{V} := \text{Ker } \omega$, the projections satisfy $\hat{\Lambda} = \text{pr}_{\mathcal{H}} \circ \widehat{B}^{-1}$, $\mathcal{H}^\perp = {}^\perp \mathcal{H} = \mathcal{V}$ and $\mathcal{V}^\perp = {}^\perp \mathcal{V} = \mathcal{H}$. When studying the associated bivector, We can always reduce ourselves to this case

Proposition
For any non-degenerate B there is a non-degenerate B' with $\text{Ker } \omega' \oplus \text{Ker } S' = TM$ whose bivector Λ' equals Λ . Hence, to study Λ one may always assume $TM = \text{Ker } \omega \oplus \text{Ker } S$.

Conversely, every bivector field $\Lambda \in \mathfrak{X}^2(M)$ arises as $-(\Lambda^2 tB^{-1})(\omega)$ for some non-degenerate B .

3. The conjugate Schouten–Nijenhuis bracket

By results due to Lichnerowicz, to know if Λ defines a Poisson or Jacobi structure we need to compute $[\Lambda, \Lambda]$.

Indeed, if $\Lambda \in \mathfrak{X}^2(M)$ is a bivector field:

- It is **Poisson** if and only if $[\Lambda, \Lambda] = 0$, and then the bracket is given by $\{f, g\} = \Lambda(df, dg)$.
- It is **Jacobi** if and only if there exists a v.f. $E \in \mathfrak{X}(M)$ such that $[\Lambda, \Lambda] = 2E \wedge \Lambda$ and $[\Lambda, E] = 0$, and then the bracket is given by $\{f, g\} = \Lambda(df, dg) + fE(g) - gE(f)$.

Definition
For $\alpha \in \Omega^a(M)$, $\beta \in \Omega^b(M)$, the conjugate of the Schouten–Nijenhuis bracket by B is $[\![\alpha, \beta]\!] := \Lambda^{a+b-1} t\bar{B}\Big(\Big[\Lambda^a t\bar{B}^{-1}(\alpha), \Lambda^b t\bar{B}^{-1}(\beta)\Big]\Big).$

With the conjugate bracket, the Poisson and Jacobi conditions on Λ , can be equivalently written, respectively, as

$$[\![\omega, \omega]\!] = 0 \quad \text{or} \quad [\![\omega, \omega]\!] = -2\alpha \wedge \omega \quad \text{and} \quad [\![\omega, \alpha]\!] = 0,$$

for some $\alpha \in \Omega^1(M)$, and then $E = t\bar{B}^{-1}(\alpha)$.

4. Horizontal and vertical distributions

Let B be a non-degenerate 2-covariant tensor and let ω and S denote, respectively, its skew-symmetric and symmetric parts. Assume that $TM = \mathcal{H} \oplus \mathcal{V}$, with $\mathcal{H} := \text{Ker } S$ and $\mathcal{V} := \text{Ker } \omega$ vector subbundles.

Theorem
For any $X, Y, Z \in \mathfrak{X}(M)$ we have $\frac{1}{2}[\![\omega, \omega]\!](X, Y, Z) = -d\omega(X_{\mathcal{H}}, Y_{\mathcal{H}}, Z_{\mathcal{H}}) - S([X_{\mathcal{H}}, Y_{\mathcal{H}}], Z_{\mathcal{V}}) - S([Y_{\mathcal{H}}, Z_{\mathcal{H}}], X_{\mathcal{V}}) - S([Z_{\mathcal{H}}, X_{\mathcal{H}}], Y_{\mathcal{V}}),$ where $X_{\mathcal{H}} = \text{pr}_{\mathcal{H}}(X)$ and $X_{\mathcal{V}} = \text{pr}_{\mathcal{V}}(X)$.

The non-integrability of $\mathcal{H}$ and $\mathcal{V}$ is measured by the vector-valued 2-forms $\text{curv}_{\mathcal{H}} \in \Omega^2(\mathcal{H}; \mathcal{V})$ and $\text{curv}_{\mathcal{V}} \in \Omega^2(\mathcal{V}; \mathcal{H})$:

$$\text{curv}_{\mathcal{H}}(X_{\mathcal{H}}, Y_{\mathcal{H}}) := \text{pr}_{\mathcal{V}}[X_{\mathcal{H}}, Y_{\mathcal{H}}], \quad \text{curv}_{\mathcal{V}}(X_{\mathcal{V}}, Y_{\mathcal{V}}) := \text{pr}_{\mathcal{H}}[X_{\mathcal{V}}, Y_{\mathcal{V}}].$$

Pairing $\text{curv}_{\mathcal{H}}$ with S defines the tensor $\iota_{\text{curv}_{\mathcal{H}}} S \in \Gamma(\Lambda^2 \mathcal{H}^* \otimes \mathcal{V}^*)$, given by $(\iota_{\text{curv}_{\mathcal{H}}} S)(H_1, H_2, V_1) := S(\text{curv}_{\mathcal{H}}(H_1, H_2), V_1)$.

Theorem
Let $B = \omega + S$ be non-degenerate such that $TM = \mathcal{H} \oplus \mathcal{V}$. Then: $\frac{1}{2}[\![\omega, \omega]\!] = -(\text{d}\omega) _{\mathcal{H}} - \iota_{\text{curv}_{\mathcal{H}}} S.$ As S is symmetric it can be diagonalized locally as $S = \lambda_k \tau_k \otimes \tau_k$ and then $\frac{1}{2}[\![\omega, \omega]\!] = -(\text{d}\omega) _{\mathcal{H}} + \sum_k \lambda_k (\text{d}\tau_k) _{\mathcal{H}} \wedge \tau_k$.

5. Poisson and Jacobi bivector fields

Assume that $TM = \mathcal{H} \oplus \mathcal{V}$, with $\mathcal{H} := \text{Ker } S$ and $\mathcal{V} := \text{Ker } \omega$. Now, we can use the last theorems to characterize the conditions for a non-degenerate 2-covariant tensor field B to be Poisson or Jacobi.

Theorem 1 (Poisson)
The bivector field Λ defines a Poisson structure iff the following conditions hold $\text{d}\omega$ vanishes on horizontal vectors, so $(\text{d}\omega) _{\mathcal{H}} = 0$. - The horizontal distribution $\mathcal{H}$ is integrable.

Theorem 2 (Jacobi)
Assume that $TM = \mathcal{H} \oplus \mathcal{V}$, where $\mathcal{H} := \text{Ker } S$ and $\mathcal{V} := \text{Ker } \omega$ are vector subbundles. Then, $\Lambda = -(\Lambda^2 tB^{-1})(\omega)$ can be completed to a Jacobi structure iff the following conditions hold - There exists a 1-form $\bar{\alpha} \in \mathcal{H}^*$ satisfying $(\text{d}\omega) _{\mathcal{H}} = \bar{\alpha} \wedge (\omega _{\mathcal{H}})$, - There exists a v.f. $R \in \Gamma(\mathcal{V})$ s.t. $\text{curv}_{\mathcal{H}}(X_{\mathcal{H}}, Y_{\mathcal{H}}) = -\omega(X_{\mathcal{H}}, Y_{\mathcal{H}})R$, - If $\alpha = \bar{\alpha} + t\bar{B}(R)$ and $X_\alpha = t\bar{B}^{-1}(\alpha)$, we have $(\mathcal{L}_{X_\alpha} \omega) _{\mathcal{H}} = (\iota_{X_\alpha} \text{d}\omega - \text{d}\bar{\alpha}) _{\mathcal{H}} = 0 \quad \text{and} \quad [X_\alpha, \mathcal{H}] \subseteq \mathcal{H}.$The Jacobi structure is $\Lambda = -(\Lambda^2 tB^{-1})(\omega)$ and $E = X_\alpha = t\bar{B}^{-1}(\bar{\alpha}) + R$.

6. Notable examples

Doublets. For a doublet (τ, ω) , set $B = \tau \otimes \tau + \omega$. It is non-degenerate $\iff$ the class is maximal ($= \dim M$), since $\text{Ker } B = \text{Ker } \tau \cap \text{Ker } \omega$. A **Reeb** field ($\iota_R \tau = 1$, $\iota_R \omega = 0$) exists if class is odd; a **Liouville** field ($\iota_\Delta \omega = \tau$) if class is even.

Theorem (even class, $\dim M = 2n$)
$B = \omega + \tau \otimes \tau$ is Poisson $\iff \text{d}\omega = 0$, and is Jacobi $\iff \text{d}\omega = \alpha \wedge \omega$ for some closed 1-form α .

Theorem
With $\mathcal{H} = \text{Ker } \tau$ and Reeb field R : the tensor $B = \omega + \tau \otimes \tau$ is Poisson $\iff \mathcal{H}$ is integrable ($\tau \wedge \text{d}\tau = 0$) and $\text{d}\omega \wedge \tau = 0$. Moreover, B is Jacobi $\iff$ there exist $\lambda \in \mathcal{C}^\infty(M)$ and $\bar{\alpha} \in \Gamma(\mathcal{H}^*)$ such that $\text{d}\omega \wedge \tau = \bar{\alpha} \wedge \omega \wedge \tau$, $\text{d}\tau \wedge \tau = \lambda(\omega \wedge \tau)$, and, for $X_\alpha = t\bar{B}^{-1}(\bar{\alpha}) - \lambda R$, $(\iota_{X_\alpha} \text{d}\tau - \text{d}\lambda) \wedge \tau = 0$ and $(\mathcal{L}_{X_\alpha} \omega) \wedge \tau = 0$. In this case, the Jacobi pair is $\Lambda = -(\Lambda^2 tB^{-1})(\omega)$ and $E = X_\alpha$.

A single formula reproduces every classical case; here $\frac{1}{2}[\![\omega, \omega]\!]$ is evaluated for each B :

Geometry	Tensor B	$\frac{1}{2}[\![\omega, \omega]\!]$	Structure
Symplectic $\text{d}\omega = 0$	ω	$-\text{d}\omega = 0$	Poisson
LCS $\text{d}\omega = -\theta \wedge \omega$	ω	$\theta \wedge \omega$	Jacobi, $E = t\omega^{-1}(-\theta) = \Delta$
Contact $\eta \wedge (\text{d}\eta)^n \neq 0$	$\text{d}\eta + \eta \otimes \eta$	$\eta \wedge \text{d}\eta$	Jacobi, $E = -R$
Cosymplectic $\text{d}\omega = \text{d}\tau = 0$	$\omega + \tau \otimes \tau$	$\text{d}\tau \wedge \tau = 0$	Poisson
Partially cosympl. $\text{d}\omega = 0$	$\omega + \tau \otimes \tau$	$\text{d}\tau \wedge \tau$	Jacobi*
Cocontact $(\text{d}\eta)^n \wedge \eta \wedge \tau \neq 0$	$\text{d}\eta + \eta \otimes \eta + \tau \otimes \tau$	$\eta \wedge \text{d}\eta$	Jacobi, $E = -R$

*Jacobi provided $\text{d}\tau \wedge \tau = -\alpha \wedge \omega$ and $[\![\omega, \alpha]\!] = 0$.

Almost cosymplectic of order p. On a $(2n+p)$ -manifold, $(\omega, \tau_1, \dots, \tau_p)$ with $\omega^n \wedge \tau_1 \wedge \dots \wedge \tau_p \neq 0$ gives $B = \omega + \sum_k \tau_k \otimes \tau_k$ and $\text{curv}_{\mathcal{H}}(H_1, H_2) = \sum_k (\text{d}\tau_k)(H_1, H_2) R_k$.

Proposition
$\Lambda = -(\Lambda^2 tB^{-1})(\omega)$ is Poisson $\iff (\text{d}\omega) _{\mathcal{H}} = 0$ and $(\text{d}\tau_k) _{\mathcal{H}} = 0$ for all $k = 1, \dots, p$; it defines a Jacobi structure $\iff$ there exist $\bar{\alpha} \in \Gamma(\mathcal{H}^*)$ and $\lambda_1, \dots, \lambda_p \in \mathcal{C}^\infty(M)$ such that $(\text{d}\omega) _{\mathcal{H}} = \bar{\alpha} \wedge (\omega _{\mathcal{H}})$, $(\text{d}\tau_k) _{\mathcal{H}} = \lambda_k (\omega _{\mathcal{H}})$ for all k , and $X_\alpha = t\bar{B}^{-1}(\bar{\alpha}) - \sum_{k=1}^p \lambda_k R_k$ satisfies $(\iota_{X_\alpha} \text{d}\tau_k) _{\mathcal{H}} = (\text{d}\lambda_k) _{\mathcal{H}}$ for all k and $(\mathcal{L}_{X_\alpha} \omega) _{\mathcal{H}} = 0$. The Jacobi structure is then $(\Lambda, E = X_\alpha)$.

Fat bundles. For an A -fat bundle $\pi: P \rightarrow M$ with connection α and a metric g on $\mathcal{V} = \text{Ker } T\pi$, the tensor $B_\mu = \mu \circ \omega_\alpha + g$ ($\mu \in A$) is non-degenerate with $TP = \mathcal{V} \oplus \text{Ker } \alpha$. The Bianchi identity gives $(\text{d}(\mu \circ \omega_\alpha))|_{\mathcal{H}} = 0$, so $\frac{1}{2}[\![\mu \circ \omega_\alpha, \mu \circ \omega_\alpha]\!] = -\iota_{\text{curv}_{\mathcal{H}}} g$.

Proposition
$(\mathcal{H}, \mu \circ \omega_\alpha)$ is a symplectic vector bundle. The bivector Λ_μ never defines a Poisson structure on P ; it completes to a Jacobi structure $\iff \omega_\alpha$ (i.e. $\text{curv}_{\mathcal{H}}$) has 1-dimensional image, and the structure is then independent of $\mu \in A$ up to a scalar.

Corollary
Every fat $U(1)$ -bundle carries a canonical induced Jacobi structure: it is a contact manifold, with the Jacobi structure of the contact form α .

7. Concluding remarks and outlook

- The obstruction for a non-degenerate 2-covariant tensor to be Poisson/Jacobi is $\frac{1}{2}[\![\omega, \omega]\!] = -(\text{d}\omega)|_{\mathcal{H}} - \iota_{\text{curv}_{\mathcal{H}}} S$.
- One intrinsic formula recovers the symplectic, LCS, cosymplectic, contact and cocontact brackets, with new results for fat bundles and order- p almost cosymplectic structures.
- Outlook:** unify the associated dynamics; relate B to G -structures; higher Poisson structures in field theory; degenerate B and Dirac–Jacobi structures.

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